

DA-Nav: Direction-Aware City-Scale Vision-Language Navigation

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Abstract—City-scale outdoor navigation is currently hindered by the heavy reliance on dense maps or costly navigation supervision. In this work, we introduce a novel paradigm for leveraging directional instructions from commercial navigation tools (e.g., Google Maps). To bridge the gap between commercial instructions and executable navigation actions, while mitigating long-horizon error accumulation through robust trajectory recovery, we propose DA-Nav, a Direction-Aware vision-language Navigation framework that reformulates navigation as a discrete spatial grounding problem on the egocentric 2D image plane. To achieve trajectory recovery, DA-Nav employs a Chain-of-Thought (CoT) reasoning process encompassing deviation assessment, action prediction, and target grid selection. We further introduce ReDA, a dataset that provides direction-aware instructions and recovery trajectories to enhance spatial grounding and support CoT recovery reasoning. Extensive experiments in CARLA demonstrate that DA-Nav achieves a high success rate of 56.16% in unseen urban environments, outperforming existing State-of-The-Art (SoTA) methods while maintaining a substantially stronger recovery capability. Furthermore, without fine-tuning, DA-Nav seamlessly adapts to both quadruped and humanoid robots, enabling stable kilometer-scale closed-loop outdoor navigation in complex real-world environments.

Index Terms—Vision-Based Navigation, Data Sets for Robot Learning, Spatial Grounding, Trajectory Recovery, Sim-to-Real Transfer

I. INTRODUCTION

City-scale outdoor mobile robots hold significant potential for applications such as logistics delivery, automated inspection, and public services. However, reliable long-horizon urban navigation remains challenging. Existing approaches typically rely on Simultaneous Localization and Mapping (SLAM) and global path planning [1], [2]. In city-scale

scenarios, however, such methods often incur substantial mapping and maintenance costs and remain sensitive to dynamic environmental changes. To reduce the reliance on dense maps, recent studies have explored navigation using coarse-grained guidance signals such as natural language instructions, GPS waypoints, and semantic landmarks [4]–[9]. Nevertheless, these approaches usually require dense navigation supervision, such as fine-grained language instruction and landmark annotations, or manually collected trajectories, which limits scalability in city-scale deployments.

To address these limitations, we investigate a new paradigm for city-scale navigation by leveraging instructions from commercial navigation tools (e.g., Google Maps, Amap), formulating it as a Direction-Aware Vision-Language Navigation (VLN) task. Compared with manually constructed dense navigation supervision, commercial navigation tools already provide mature global path planning and real-time local directional guidance. However, such instructions are designed for human cognition and only contain coarse-grained directional information (e.g., “turn right in 50 meters”), making them unsuitable for direct robotic execution. Therefore, the first challenge in this paradigm is how to ground sparse directional instructions with egocentric observations into locally executable motion behaviors. Recent development of Vision-Language Models (VLM) [22], [25] and Vision-Language-Action (VLA) models [26], [27] demonstrate the potential for solving the cross-modal alignment problem. Moreover, during long-horizon navigation, perception and control errors inevitably accumulate over time, causing the robot to gradually deviate from the intended path. Consequently, robust recovery from trajectory deviations is another critical challenge.

To this end, we propose a **Direction-Aware Navigation (DA-Nav)** method based on VLM for city-scale navigation. Unlike conventional continuous trajectory regression methods [8]–[10], DA-Nav reformulates navigation as a vision-language-conditioned discrete spatial grounding problem on the 2D image plane. As illustrated in Fig. 1, we construct a discrete spatial grid over egocentric visual observations and leverage the spatial grounding capability of VLMs to predict local navigation targets directly on the image plane. Furthermore, DA-Nav introduces a structured sequential reasoning process that decomposes navigation into a series of decision steps, including deviation assessment, action prediction, and image-plane target selection. Specifically, the model first determines whether trajectory deviation has occurred, then predicts the corresponding navigation or corrective action, and finally generates the target grid sequence on the image

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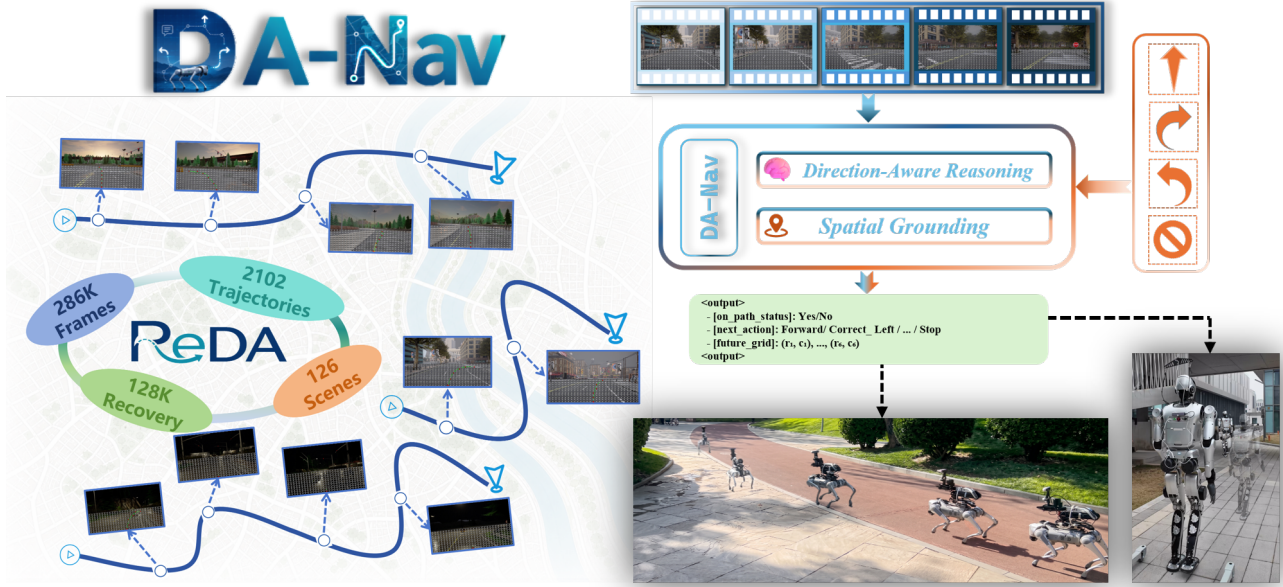


Fig. 1. Overview of DA-Nav. Trained on the ReDA dataset, our policy reformulates outdoor navigation as a vision-language-conditioned discrete spatial grounding problem on the egocentric image plane. The model autoregressively predicts structured decisions and image-plane trajectories, achieving robust zero-shot sim-to-real deployment across diverse legged robots.

plane. Implemented in a Chain-of-Thought (CoT) formulation [11]–[15], this structured decision process enables trajectory recovery during long-horizon navigation.

In addition, enabling direction-aware navigation and trajectory recovery requires training data with directional instructions and recovery behaviors. However, the instructions in existing outdoor navigation datasets are predominantly landmark-driven rather than direction-aware instructions [4], [5], [18]. Besides, current datasets are limited to expert trajectories and lack supervision for recovery behaviors [9], [34], [35]. Therefore, we introduce **ReDA** dataset, including both **Recovery** trajectories and **Direction-Aware** instructions. Specifically, we develop an automated data pipeline to generate directional instructions across 2102 trajectories within 126 scenes in the CARLA simulator [16]. Besides, we actively inject perturbations to generate deviated trajectories and construct corresponding CoT reasoning steps for training recovery capability.

To validate the effectiveness of the proposed approach, we conducted extensive closed-loop evaluations in the CARLA simulator. DA-Nav achieved a navigation success rate of 56.16% in unseen urban environments, outperforming other State-of-The-Art (SoTA) methods. Furthermore, the learned policy achieves zero-shot transfer to real-world robots, including a Unitree Go2 quadruped robot and a Leju Kuavo-V humanoid robot, without any real-world fine-tuning, enabling stable kilometer-scale closed-loop outdoor navigation. Our main contributions are summarized as follows:

- We propose DA-Nav, a VLM-based long-horizon outdoor navigation model driven by direction-aware instructions. It performs discrete image-plane spatial grounding and CoT reasoning for trajectory recovery.
- We introduce ReDA, a dataset comprising direction-aware instructions and trajectory recovery annotations,

enabling robust learning under Out-Of-Distribution (OOD) deviation states.

- DA-Nav achieves SoTA performance in CARLA. Notably, it demonstrates exceptional zero-shot sim-to-real transfer, achieving stable long-horizon navigation over 1.2 km in real-world environments.

II. RELATED WORK

A. Outdoor Navigation in Unstructured Environments

To bypass the prohibitive mapping costs and dynamic sensitivity inherent in conventional SLAM-based systems [1], [2], recent approaches have explored high-level guidance such as GPS waypoints, topological graphs, and semantic landmarks [6], [8], [9]. However, maintaining reliable localization and semantic consistency in large unstructured outdoor environments remains challenging due to GPS multi-path errors and dynamic scene variations. While conventional VLN methods rely on fine-grained instructions to describe specific paths [4], such dense supervision is unrealistic for city-scale deployment. In contrast, commercial navigation tools provide lightweight coarse-grained guidance for long-horizon navigation, but such instructions lack the precision required for direct robotic control. DA-Nav addresses this challenge by grounding sparse directional guidance into reliable local spatial decisions.

B. Vision-Language Models for Embodied Navigation

The strong semantic understanding and cross-modal alignment capabilities of VLMs have driven a paradigm shift in embodied navigation [19]–[21], [25]. While end-to-end VLA models such as RT-2 [26] and OpenVLA [27] exhibit impressive generalization, modeling continuous spatial actions in an autoregressive manner remains challenging for stable long-horizon execution [22], [28]. Recent studies suggest

that VLMs are more effective at reasoning over 2D visual representations than regressing 3D waypoints [22]–[24]. This indicates that VLM-based navigation instability largely stems from a discrepancy between model capability and action representation. Motivated by this, DA-Nav reformulates navigation planning as a vision-language-conditioned discrete spatial grounding problem. By projecting candidate targets onto a discretized egocentric image grid and performing decision-making directly on the image plane, DA-Nav aligns the action space with the visual reasoning characteristics of VLMs, thereby mitigating spatial hallucination and improving navigation stability.

C. Closed-Loop Error Recovery and Reasoning

Most existing vision-based navigation policies are trained via behavior cloning on nominal expert-only demonstrations [6], [10]. While effective in offline settings, this paradigm inevitably suffers from covariate shift during closed-loop execution [29]–[31]. Minor execution errors can quickly push the agent into OOD states, leading to compounding deviations and eventual failure. Interactive approaches such as DAgger [29] mitigate this issue by querying expert supervision online, but they require expensive simulator-in-the-loop setups, making them difficult to scale to large VLM-based systems. Recent works have explored structured reasoning and decision decomposition to improve interpretability in embodied systems [32], [33]. However, most existing approaches focus primarily on high-level task planning, with limited consideration of low-level execution assessment and recovery under deviation states. In contrast to prior approaches that primarily rely on expert-only demonstrations or high-level planning supervision, DA-Nav incorporates corrective supervision and direction-aware CoT reasoning for robust long-horizon closed-loop execution.

III. TASK DEFINITION AND REDA DATASET

A. Direction-Aware VLN

In our task, at each time step t , the robot receives a sequence of egocentric RGB observations $\mathbf{O}_t = (o_{t-k}, \dots, o_t)$ with $k = 4$, together with a discrete directional instruction $I_t \in \{\text{FORWARD}, \text{TURN_LEFT}, \text{TURN_RIGHT}, \text{STOP}\}$ provided by commercial navigation tools. Given the multimodal context $C_t = \{\mathbf{O}_t, I_t\}$, the goal is to learn a policy π_θ that predicts a feasible local navigation trajectory. Unlike conventional approaches that directly regress continuous 3D waypoints, we reformulate navigation as a vision-language-conditioned discrete spatial grounding problem on the egocentric image plane. Future trajectories are represented as discretized candidate locations on the image plane.

B. Dataset Description

As summarized in Table I, existing navigation datasets predominantly rely on fine-grained language instructions, metric action space, and expert-only demonstrations. While effective for supervised navigation learning, these designs remain inadequate for robust long-horizon closed-loop navigation. Dense natural language supervision introduces unnecessary

TABLE I
COMPARISON OF REDA WITH REPRESENTATIVE VISION-LANGUAGE NAVIGATION DATASETS

Dataset	Environment	Depth	Instruction Type	Action Space
R2R [34]*	Indoor	×	Fine-grained Description	Graph Nodes
Touchdown [5]*	Outdoor	×	Fine-grained Description	Graph Nodes
VLN-CE [35]*	Indoor	✓	Fine-grained Description	3D Control
Talk2Nav [18]*	Outdoor	×	Fine-grained Description	3D Waypoints
LM-Nav [4]*	Outdoor	×	Landmark-based	Graph Nodes
GNM [36]*	Mixed	×	Image Goal	2D Waypoints
CityWalker [9]*	Outdoor	×	Path-level	3D Waypoints
NaVid [7]*	Indoor	×	Path-level	Discrete Commands
NaVILA [37]*	Mixed	✓	Mixed	Discrete Commands
ReDA (Ours)[†]	Outdoor	×	Direction-based	2D Image Grid

Note: All datasets use RGB input. * expert-only; [†] expert + recovery data.

semantic complexity, continuous metric-space grounding with VLMs remains challenging, and expert-only trajectories lack recovery behaviors for OOD states.

To address these limitations, we introduce ReDA, which jointly integrates directional instructions, discretized 2D image-space grounding, and trajectory recovery data generation. By grounding navigation targets directly onto an egocentric image grid and explicitly incorporating recovery trajectories from deviated states during data collection, ReDA provides a dataset tailored for training robust closed-loop spatial reasoning and autonomous deviation recovery. The two core components of ReDA, namely the *Egocentric Grid Trajectory Representation* and the *Direction-Aware Recovery Data Generation*, are detailed in the following subsections.

C. Egocentric Grid Trajectory Representation

To instantiate this direction-aware spatial selection problem, we construct a discrete grid on the egocentric RGB image plane and define the valid spatial region as:

$$G = \{(r, c) \mid r \in [13, 23], c \in [0, 28]\} \subset \mathbb{Z}^2, \quad (1)$$

where r and c denote the row and column indices, corresponding to forward and lateral directions, respectively. The valid region excludes sky and distant background areas, focusing on traversable regions in front of the robot. Based on this representation, we discretize the future trajectory into a sequence of grid locations with a horizon length of $L = 6$:

$$\mathbf{P}_t = (\mathbf{g}_{t+1}, \dots, \mathbf{g}_{t+L}), \quad \mathbf{P}_t \in \mathbb{R}^{L \times 2}. \quad (2)$$

This sequence represents the robot motion over the next 3 seconds sampled at 2 Hz. This representation constrains trajectory prediction to discrete image-space locations.

Building upon this grid representation, we further construct a structured CoT decision sequence $Y_t = (s_t, c_t, \mathbf{P}_t)$. Here, $s_t \in \{\text{Yes}, \text{No}\}$ serves to explicitly evaluate whether the robot has deviated from the reference path; $c_t \in \{\text{FORWARD}, \text{TURN_LEFT}, \text{TURN_RIGHT}, \text{CORRECT_LEFT}, \text{CORRECT_RIGHT}, \text{STOP}\}$ serves as the CoT action; and $\mathbf{P}_t$ denotes the predicted spatial grid sequence.

D. Direction-Aware Recovery Data Generation

To expand the state distribution, we propose an automated pipeline in the CARLA simulator governed by a three-state Finite State Machine (FSM): STABLE, DRIFTING, and

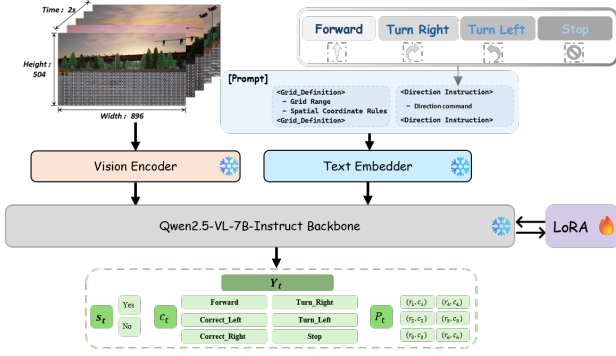


Fig. 2. Overview of the DA-Nav architecture. Multimodal inputs are processed by frozen encoders and passed to a LoRA-finetuned Qwen2.5-VL-7B backbone. The model autoregressively outputs a structured CoT decision sequence consisting of state assessment, corrective action, and discrete future trajectories on the image grid.

RECOVERING. In the **STABLE** state, the agent tracks the expert trajectory. To induce OOD states, steering perturbations trigger the **DRIFTING** state. Once the lateral error exceeds a safety threshold ($e_y \geq 0.35$ m), the FSM switches to the **RECOVERING** state. During recovery, the agent adopts an adaptive tracking strategy in which the look-ahead distance l_d is proportional to e_y , enabling smooth trajectory correction. To avoid learning unstable behaviors, frames from the **DRIFTING** state are discarded.

For the retained frames, we synthesize discrete navigation instructions based on the agent’s turning states and motion status, including **FORWARD**, **TURN_LEFT**, **TURN_RIGHT**, and **STOP**. The ground-truth deviation state s_t^* records the active FSM mode, while the corrective action c_t^* is determined by the sign of the lateral deviation e_y during recovery. Finally, future expert 3D waypoints are smoothed via cubic spline interpolation, projected onto the egocentric image plane using the calibrated camera projection model, and discretized into trajectory labels P_t^* . The resulting dataset contains approximately 286k sequential samples, including 158k expert navigation frames and 128k recovery frames.

IV. METHODOLOGY

As shown in Fig. 2, DA-Nav takes temporal egocentric observations and sparse directional instructions as input and reformulates long-horizon navigation as a vision-language-conditioned discrete spatial grounding problem. Specifically, a structured VLM policy autoregressively predicts a CoT decision sequence together with discrete image-plane navigation targets, which are subsequently projected into the robot body frame and executed through a furthest-point control interface for robust outdoor navigation.

A. DA-Nav

We instantiate the policy π_θ using the pre-trained Qwen2.5-VL-7B-Instruct backbone. At each timestep, the model takes a multimodal context C_t as input. The visual history O_t is encoded by a frozen visual encoder and projected into the language embedding space via a lightweight adapter, allowing seamless integration with the textual instruction I_t .

To improve spatial reasoning under trajectory deviations, we adopt a structured prompting strategy with output constraints. First, we inject spatial grounding constraints by encoding the discretized image grid G in the prompt, establishing the correspondence between grid indices and spatial locations. Second, we enforce logical decomposition by structuring the output generation as a sequential decision process, where the model first performs state assessment, then selects an action, and finally predicts the trajectory. This formulation encourages the VLM to explicitly evaluate the navigation state s_t and infer a high-level action c_t before predicting the spatial trajectory P_t , improving the consistency between CoT decisions and spatial trajectory prediction.

The policy is trained as an autoregressive generation process over the decision sequence Y_t . Given the context C_t , the policy factorizes as:

$$\pi_\theta(Y_t | C_t) = \prod_{j=1}^{|Y_t|} P_\theta(y_t^{(j)} | y_t^{(<j)}, C_t), \quad (3)$$

π_θ is optimized using a standard next-token prediction objective with supervision from the ReDA dataset.

To adapt the pre-trained VLM without catastrophic forgetting, we apply LoRA to the attention blocks, keeping the visual encoder and base LLM weights frozen. This parameter-efficient strategy enables the model to internalize corrective patterns from the ReDA dataset while preserving its inherent cross-modal generalization capabilities.

B. Image-Plane-to-Body-Frame Spatial Grounding

To bridge the gap between discrete image-space predictions and continuous physical execution, the geometric control interface first transforms the predicted pixel grid sequence P_t into a 3D continuous trajectory $W_t = (w_{t+1}, \dots, w_{t+L})$ in the robot’s local body frame, where $w_k = (x_{\text{body}}^{(k)}, y_{\text{body}}^{(k)})$, as illustrated in Fig. 3.

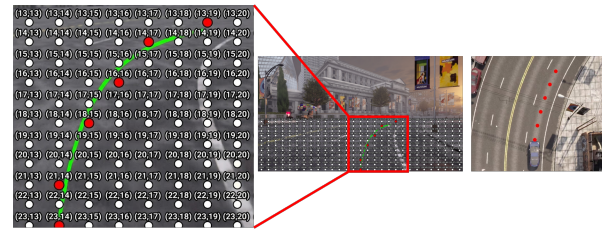


Fig. 3. Visualization of the egocentric grid representation and spatial grounding process. **Left and Middle:** The traversable region in the RGB observation is discretized into a 2D image-plane grid, where future trajectories are represented as discrete targets P_t (red dots). **Right:** The predicted targets are projected into continuous local-body-frame trajectories W_t for robot execution.

To ensure robustness across diverse outdoor environments, we employ a hybrid spatial projection strategy. When reliable depth observations are available, the predicted image-plane targets are projected into metric coordinates using the camera intrinsic matrix and local depth statistics. If depth sensing becomes unreliable, the system falls back to an Inverse

Perspective Mapping (IPM) strategy under a flat-ground assumption. This hybrid design ensures continuous and reliable spatial grounding across varying sensing conditions.

C. Furthest-Point Control Interface

Due to the latency of autoregressive VLM inference, directly tracking dense predicted trajectories may introduce high-frequency control fluctuations during closed-loop execution. To overcome this, we adopt a furthest-point target pursuit strategy. Instead of tracking all predicted waypoints, we select the furthest waypoint in the prediction horizon as the control target, i.e., $\mathbf{w}_{\text{target}} = \mathbf{w}_{t+L}$. Using the furthest predicted waypoint as the control target provides a temporally smoother furthest-point signal and reduces sensitivity to short-term prediction noise.

Guided by this furthest-point target, an adaptive low-level controller computes the final motion commands. A heading-error-driven controller computes the angular velocity:

$$\omega_z = \text{sat}_{\omega_{\max}}(k_{\text{steer}} \cdot \arctan 2(y_{\text{target}}, x_{\text{target}})), \quad (4)$$

where $\text{sat}_{\omega_{\max}}(\cdot)$ denotes saturation to enforce actuator limits. Furthermore, the linear velocity is adaptively adjusted based on the steering magnitude:

$$v_{\text{target}} = \begin{cases} v_{\text{nom}}, & |\omega_z| \leq \omega_{\text{th}} \\ v_{\text{low}}, & |\omega_z| > \omega_{\text{th}}. \end{cases} \quad (5)$$

This formulation automatically restricts aggressive steering via limits and decelerates the robot during sharp maneuvers, improving kinematic stability and enabling smooth platform adaptation during closed-loop deployment.

V. EXPERIMENTS

A. Experimental Setup

To comprehensively evaluate our method, we compare DA-Nav against five representative baselines. For end-to-end foundation models, we select *CityWalker* [9], a Transformer-based waypoint regression policy, and *ViNT* [6], a goal-conditioned topological navigation model. To assess large vision-language capabilities, we include *NaVid* [7], which directly deduces physical actions from video streams, alongside the unmodified *Zero-shot Qwen2.5-VL-7B* [17] to isolate the impact of our task-specific fine-tuning. Finally, we compare against *NaVILA* [37], a hierarchical VLA system that employs a VLM for mid-level navigation instruction generation coupled with low-level execution.

To enable consistent closed-loop evaluation across diverse navigation paradigms, all baselines are evaluated using their officially released pre-trained weights without environment-specific fine-tuning. We performed standardized interface adaptations for closed-loop deployment: routing priors are provided as moving-horizon local goal images (5 m look-ahead) for *ViNT* or deterministic sparse waypoints for *CityWalker*. For VLM-based action models (*NaVid* and *NaVILA*), identical egocentric visual histories and unified low-level proportional controllers are applied. Goal termination relies solely on their visual recognition of semantic destinations (e.g., a distinct red vehicle). Finally, *Zero-shot*

TABLE II
QUANTITATIVE COMPARISON ON 239 TRAJECTORIES

Method	SR (%) ↑	RC (%) ↑	SPL ↑	DF ↓	CSR (%) ↑
CityWalker [9]	38.08	70.79	37.48	2.96	30.73
ViNT [6]	51.88	74.44	51.33	1.25	23.54
NaVid [7]	20.50	66.33	20.50	0.61	4.89
NaVILA [37]	22.59	79.49	22.59	0.39	4.90
Zero-shot Qwen2.5-VL [17]	11.30	43.21	11.30	3.42	42.76
DA-Nav (Ours)	59.00	77.82	58.66	1.85	98.15

Qwen2.5-VL is evaluated as a direct replacement within our framework, maintaining identical system settings.

For evaluation, in addition to standard embodied navigation metrics including Success Rate (SR), Route Completion (RC), and Success weighted by Path Length (SPL), we note that these conventional metrics fail to adequately capture the model’s recovery capability under trajectory deviations. To more precisely evaluate autonomous correction, we additionally introduce two recovery-oriented metrics: Deviation Frequency (DF) and Correction Success Rate (CSR). A deviation event is counted once when the lateral error first exceeds the safety threshold after returning to a nominal state. Let E denote the set of such deviation events where the lateral error e_y exceeds a safety threshold δ_{safe} (e.g., 0.35 m), D_{total} the total traveled distance (in meters), and $E_{\text{success}} \subseteq E$ the subset of events successfully reconverging to $e_y < \delta_{\text{safe}}$ without collision. These metrics are formally defined as:

$$\text{DF} = \frac{|E|}{D_{\text{total}}} \times 100, \quad \text{CSR} = \frac{|E_{\text{success}}|}{|E|} \times 100\%. \quad (6)$$

B. Quantitative Benchmark in Simulation

All simulation experiments are conducted in CARLA 0.9.15. Towns 01–05 and 10HD are used for training and in-domain evaluation, while Towns 06, 07, and 15 are reserved for zero-shot generalization. To increase environmental diversity, dynamic traffic, pedestrians, and randomized weather and lighting conditions are introduced during both data collection and evaluation.

We evaluate all methods on the same set of 239 long-horizon closed-loop trajectories across seen and unseen towns. Quantitative results are reported in Table II, with per-town generalization results shown in Fig. 5. DA-Nav achieves the best overall performance, obtaining an SR of 59.00%, SPL of 58.66, and CSR of 98.15%. Moreover, its SR decreases by only 7.28% when transferred from seen to unseen environments, demonstrating robust spatial generalization and trajectory recovery under distribution shifts.

In contrast, existing baselines exhibit distinct failure modes during long-horizon closed-loop execution. *CityWalker*, based on continuous coordinate regression, lacks explicit corrective mechanisms, causing accumulated lateral drift and frequent unrecoverable collisions (CSR = 30.73%). Although *ViNT* achieves relatively strong route completion, it remains sensitive to perceptual aliasing in repetitive urban scenes. In 24 failure cases, the agent exceeded 120% of the optimal path length without reaching the goal. End-to-end VLM methods (*NaVid*, *NaVILA*) exhibit a different limitation: despite relatively high route completion rates,

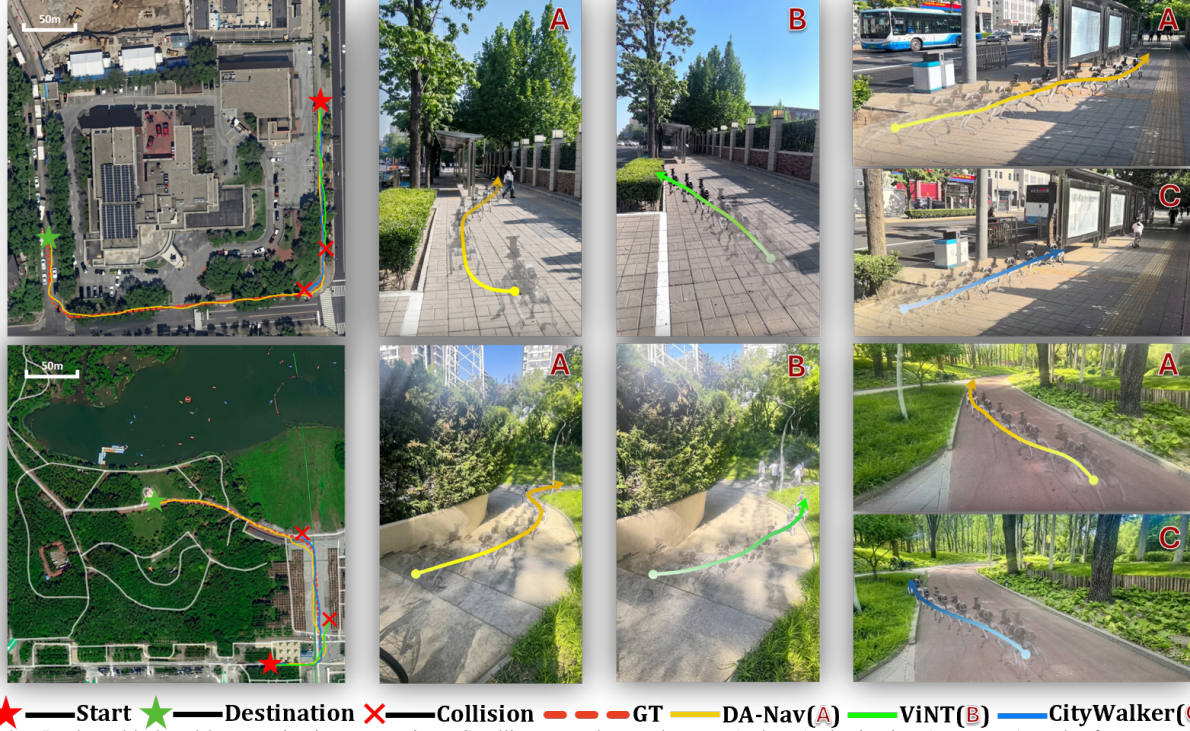


Fig. 4. Real-world closed-loop navigation comparison. Satellite maps denote the start (red star), destination (green star), and reference path (red dashed line). Third-person views visualize the execution trajectories of DA-Nav (A, yellow), ViNT (B, green), and CityWalker (C, blue). Red crosses (×) indicate collisions during execution.

both suffer from extremely weak recovery capability ($CSR \approx 5\%$) and unreliable goal termination behaviors, including overshooting and premature stopping near the destination. Although NaVILA exhibits a low deviation frequency, once significant deviations occur, the policy rarely recovers successfully. These results suggest that existing methods struggle to jointly achieve stable spatial grounding, robust deviation recovery, and reliable goal termination. By explicitly modeling CoT reasoning for decision-making and image-space spatial grounding, DA-Nav achieves substantially more robust long-horizon closed-loop navigation.

C. Ablation Studies

To validate the core components of DA-Nav, we ablate the structured reasoning process and training data distribution. Two variants are evaluated on the same 239 trajectories:

- **w/o CoT Reasoning:** Removes the structured reasoning prompt, forcing the model to directly predict final spatial targets from multimodal observations.
- **w/o Recovery Data:** Excludes all recovery samples during training, reducing the policy to pure behavior cloning on expert demonstrations.

The quantitative results for all variants are summarized in Table III. When the CoT reasoning is removed, the navigation SR drops significantly to 38.91%. More critically, the DF increases sharply to 4.30. In complex OOD scenarios (e.g., Town 15), the absence of CoT guidance leads to frequent inconsistent navigation decisions, with the local DF surging

TABLE III
ABLATION STUDIES ON 239 TRAJECTORIES

Model Variant	SR (%) ↑	RC (%) ↑	SPL ↑	DF ↓	CSR (%) ↑
w/o Recovery Data	29.71	62.01	29.68	1.31	15.46
w/o CoT Reasoning	38.91	68.37	38.91	4.30	50.11
DA-Nav (Full)	59.00	77.82	58.66	1.85	98.15

to 18.75. This indicates that structured CoT reasoning stabilizes sequential navigation decisions under complex urban layouts. Without explicit structured decision reasoning, the model becomes prone to inconsistent navigation decisions in complex urban layouts.

When the corrective data is removed, the model exhibits substantial performance degradation, achieving an SR of only 29.71%. Notably, this variant maintains a relatively low Deviation Frequency ($DF = 1.31$), indicating that it can accurately imitate expert trajectories within the nominal state distribution. However, once deviations occur, its CSR drops to only 15.46%. This strongly confirms that behavior cloning based solely on perfect demonstrations is insufficient to handle distributional shifts, whereas the proposed corrective data provides the essential kinematic recovery capability.

D. Real-World Deployment

1) **System Overview and Instruction Interface:** To validate the zero-shot sim-to-real transferability of DA-Nav, we deploy the framework on robotic platforms, including a Unitree Go2 quadruped and a Leju Kuavo-V humanoid robot, as shown in Fig. 6. The robots are equipped with a forward-

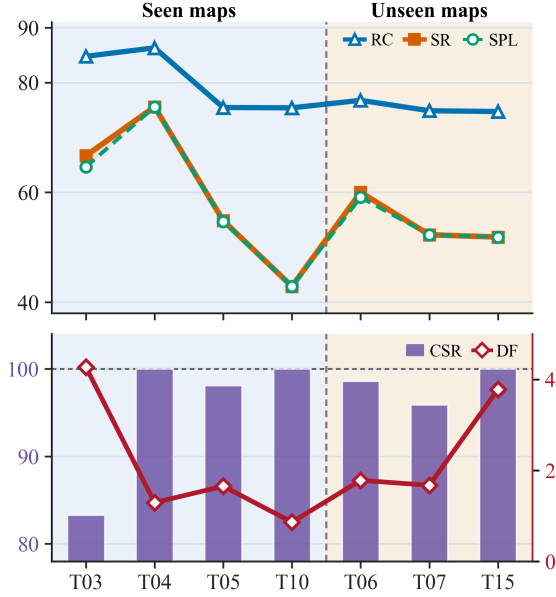


Fig. 5. Per-town evaluation of DA-Nav across seen and unseen environments. **Top:** Standard navigation metrics (RC, SR, SPL) across different CARLA towns. **Bottom:** Relationship between deviation frequency (DF, red line) and correction success rate (CSR, purple bars).

facing Intel RealSense D455 RGB camera for egocentric visual perception. Additionally, each platform is tethered to a smartphone to interface with commercial navigation tools and retrieve real-time direction-aware instructions.



Fig. 6. Robotic platforms used for real-world deployment. **Left:** Unitree Go2 quadruped robot. **Right:** Leju Kuavo-V humanoid robot. Both platforms stream egocentric RGB observations and direction-aware instructions to a remote GPU server for policy inference.

Due to the substantial computational overhead of VLMs, policy inference is executed on a remote GPU server equipped with an NVIDIA RTX 4090. The onboard platform continuously streams multimodal observations and receives structured navigation decisions through a low-latency communication interface for real-time closed-loop execution. To connect coarse-grained commercial navigation tools with local robotic control, we further develop a lightweight navigation instruction parsing module. As illustrated in the inset of Fig. 6, the module asynchronously extracts real-time path guidance and textual navigation information (e.g., “Turn left in 30 meters”) from commercial navigation tools and continuously converts them into the structured navigation instruction I_t used by DA-Nav as policy input.

TABLE IV

Method	Forward (12)	Turn Left (12)	Turn Right (12)	Avg SR (%)
CityWalker [9]	100.0	66.7	58.3	75
ViNT [6]	58.3	50.0	66.7	58.3
DA-Nav (Ours)	100.0	75.0	75.0	83.3

2) *Open-Loop Primitive Evaluation:* To isolate the execution capability of low-level navigation behaviors without the influence of accumulated errors, we first evaluate open-loop performance on basic primitives. We conduct experiments across 36 different intersections (12 cases each for straight, left, and right turns), with each case evaluated once.



Fig. 7. Real-world open-loop evaluation of navigation primitives (Forward, Turn Right, and Turn Left) across different methods. Red icons indicate failures or unsafe maneuvers. Compared with CityWalker and ViNT, DA-Nav demonstrates more stable execution and improved spatial grounding during large turning maneuvers.

As reported in Table IV and visualized in Fig. 7, baseline methods demonstrate competitive performance for straight-line navigation, but struggle severely with the large heading changes required for turning. This indicates that large heading changes pose substantial challenges for end-to-end regression-based approaches, which often lack explicit geometric constraints for stable motion generation.

In contrast, by combining structured decision prediction with image-space spatial grounding, DA-Nav demonstrates more stable and geometrically consistent turning behavior, achieving an average success rate of 83.3%.

3) *Closed-Loop Navigation in Real-World Environments:* Following the primitive validation, we evaluate full closed-loop navigation performance across two distinct real-world scenarios: urban streets and parks. For each scenario, five paths are selected (each exceeding 100m with at least one turn), and each path is repeated three times. As reported in Table V and visualized in Fig. 4, baseline methods perform poorly in real-world closed-loop settings despite having access to pre-recorded path priors. Environmental uncertainties, such as dynamic pedestrians and illumination variations, significantly degrade their performance, resulting in overall success rates of only 23.3% for CityWalker and 16.7% for ViNT. In contrast, DA-Nav achieves an overall success

TABLE V

REAL-WORLD CLOSED-LOOP NAVIGATION PERFORMANCE

Method	Urban (15 Trials)		Park (15 Trials)		Overall	
	SR (%)	RC (%)	SR (%)	RC (%)	SR (%)	RC (%)
CityWalker [9]	20.0	47.6	26.7	58.9	23.3	53.3
ViNT [6]	13.3	43.4	20.0	54.5	16.7	49.0
DA-Nav (Ours)	40.0	68.4	53.3	84.2	46.7	76.3

rate of 46.7%. By leveraging direction-aware reasoning and explicit recovery capability, DA-Nav continuously corrects trajectory deviations during execution, ensuring robust navigation under real-world uncertainties. These results highlight that reliable long-horizon navigation critically depends on direction-aware guidance and active recovery capabilities, rather than solely on passive trajectory imitation.

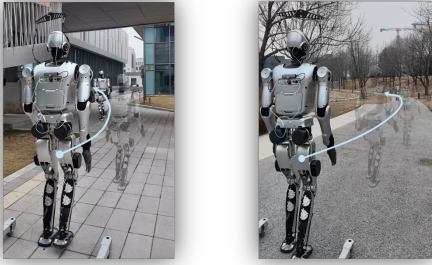


Fig. 8. Zero-shot cross-embodiment deployment on the Leju Kuavo-V humanoid. The model achieves continuous closed-loop navigation over 1.2 km without real-world fine-tuning, exhibiting stable trajectory execution (blue curves) in both (Left) urban and (Right) park environments.

4) *Cross-Embodiment Generalization:* Beyond the quadruped platform, we deploy the pre-trained DA-Nav policy without modification onto a full-scale humanoid robot (Leju Kuavo-V), as shown in Fig. 8. Without embodiment-specific fine-tuning, it achieves successful goal-directed navigation over 1.2 km in outdoor environments, demonstrating strong cross-embodiment generalization.

VI. CONCLUSION

In this paper, we present DA-Nav, a direction-aware framework that integrates CoT reasoning for trajectory recovery in robust long-horizon closed-loop navigation. It achieves stable navigation under both simulated and real-world environmental perturbations. Extensive experiments in CARLA and real-world deployments across quadruped and humanoid platforms demonstrate strong generalization, autonomous recovery capability, and cross-embodiment transferability.

Despite these advantages, the current system still relies on commercial navigation tools for high-level path guidance, which introduces practical limitations such as GPS inaccuracies, update latency, and restricted coverage in unmapped regions. Future work will focus on eliminating this dependency through autonomous exploration and lifelong topological memory construction. By enabling robots to continuously build persistent spatial representations in unknown environments, we aim to achieve robust long-horizon navigation in fully unstructured and GPS-denied scenarios.

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